

Management of Anaphylaxis in Paediatric Patients

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1) Introduction and Diagnostic Criteria

Anaphylaxis is a severe, potentially fatal, systemic allergic reaction that occurs suddenly after contact with an allergy-causing substance.

Table 1: Diagnostic criteria for Anaphylaxis (from National Institute of Allergy and Infectious Diseases)

Anaphylaxis is highly likely when any 1 of the following 3 criteria are fulfilled:	
1	Acute onset of illness (min to several hrs) with involvement of skin, mucosal tissue, or both (e.g. generalized urticaria, pruritus or flushing, swollen lips-tongue-uvula) AND ≥ 1 of the following: a) Respiratory compromise (e.g. dyspnea, wheeze-bronchospasm, stridor, reduced peak expiratory flow, hypoxaemia) b) [†]Hypotension or associated symptoms of end organ dysfunction (e.g. collapse, syncope, incontinence)
2	≥ 2 of the following that occur rapidly (min to several hrs) after exposure to a likely allergen for that patient: a) Involvement of skin-mucosal tissue (e.g. generalized urticaria, pruritus or flushing, swollen lips-tongue-uvula) b) Respiratory compromise (e.g. dyspnea, wheeze-bronchospasm, stridor, reduced peak expiratory flow, hypoxaemia) c) [†]Hypotension or associated symptoms associated symptoms of end organ dysfunction (e.g. collapse, syncope, incontinence) d) ^Δ<u>Persistent</u> gastrointestinal symptoms (e.g. crampy, abdominal pain, vomiting)
3	Hypotension after exposure to known allergen for that patient (onset min to several hrs)

[†]Hypotension defined as:

- Systolic BP (SBP) < 70 + (age x 2) (up to 90 mmHg) OR
- > 30% decrease in SBP from baseline

^ΔA single episode of vomiting with subsequent clinical improvement (i.e. patient feels better, rash not progressing) does not constitute anaphylaxis.

2) Management Principles (Diagram 1)

- **IM adrenaline 1:1000 (NEAT) 0.01 mL/kg (max 0.5 mL) is the mainstay of treatment and should not be delayed once anaphylaxis is suspected.**
- When in doubt, IM adrenaline should still be given.
- Consider sending serum tryptase if the diagnosis of anaphylaxis is unclear. This, however, should not delay the administration of IM adrenaline.
- 2nd line medications can be given after IM adrenaline but these should not not delay administration of, or be given in place of, IM adrenaline. Refer to [table 2](#) for the doses of these medications.

A) Indications for Admission (High-Risk)

- 1) Severe/protracted anaphylaxis (e.g. shock, requiring >1 dose of IM adrenaline)
- 2) Young age < 6 mths OR weight < 7.5 kg
- 3) Allergen avoidance not possible/difficult
- 4) No reliable caregiver
- 5) High risk comorbidities (e.g. severe/poorly controlled asthma, arrhythmias, systemic mastocytosis, beta-blocker use)

B) Management in Children's Emergency (CE) if Admission not indicated

- Patients without the above risk factors can be observed in the Children's Emergency for **4 – 6 hours from the time of administration of IM adrenaline** (either in hospital or before arriving in hospital if an adrenaline auto-injector is used) for a biphasic reaction.
 - During this time, a comprehensive exposure history prior to reaction should be obtained and clearly documented in the clinical notes to facilitate outpatient work-up in 4 – 6 weeks' time.
- Parents should be advised to obtain a list of ingredients (for food items where appropriate) and to bring it along to the clinic visit. Use of an adrenaline auto-injector (Epipen® or Epipen Junior® in our institution, refer to [table 3](#) for dose) should be taught, and an anaphylaxis action plan given prior to discharge.
- Adrenaline auto-injectors do ***not*** need to be given for **drug-induced anaphylaxis** as this should be flagged up on the CMIS system and a medical alert card (printed from CMIS system) given to the patient if confirmed.

3) Special consideration: Infant < 6 months old

A) Allergic reaction not satisfying anaphylaxis criteria

- Off label use of a **NON-SEDATING** antihistamine, e.g. PO cetirizine, may be considered in a child nearing 6 months of age who has significant active allergic symptoms (e.g. isolated generalized urticaria):
 - < 4 months: consult allergy
 - ≥ 4 months: PO cetirizine 1 – 2 mg once dose
- Strict avoidance of the suspected allergen should be advised. Admit patient if allergen avoidance is not possible or alternatives not readily available (e.g. cow's milk based formula).
- Anti-histamines should ***not*** be prescribed on discharge.

B) Anaphylaxis

- IM adrenaline 1:1000 (NEAT) 0.01 mL/kg remains the mainstay of treatment.
- Admit patient for observation and consult allergy team inpatient for counselling if allergen unclear or avoidance not possible.

4) Discharge Checklist for Anaphylaxis in CE

- ☐ Instructions for strict avoidance of suspected trigger(s) (*report allergy in CMIS if drug allergy suspected*)
- ☐ Anaphylaxis action plan explained and provided
- ☐ Use of adrenaline auto-injector taught (indications and technique)
- ☐ Educate regarding chance of biphasic reaction (up to 78 hours from initial reaction) and to return to CE
- ☐ Adult Epipen® or Epipen Junior® and PO antihistamine (standby for subsequent reactions)
- ☐ Clear documentation of all exposures prior to reaction in clinical records
- ☐ TCU Allergy x 4 – 6 weeks (no antihistamines at least 3 days before)

Diagram 1: Management of Allergic Reactions in Paediatric Patients

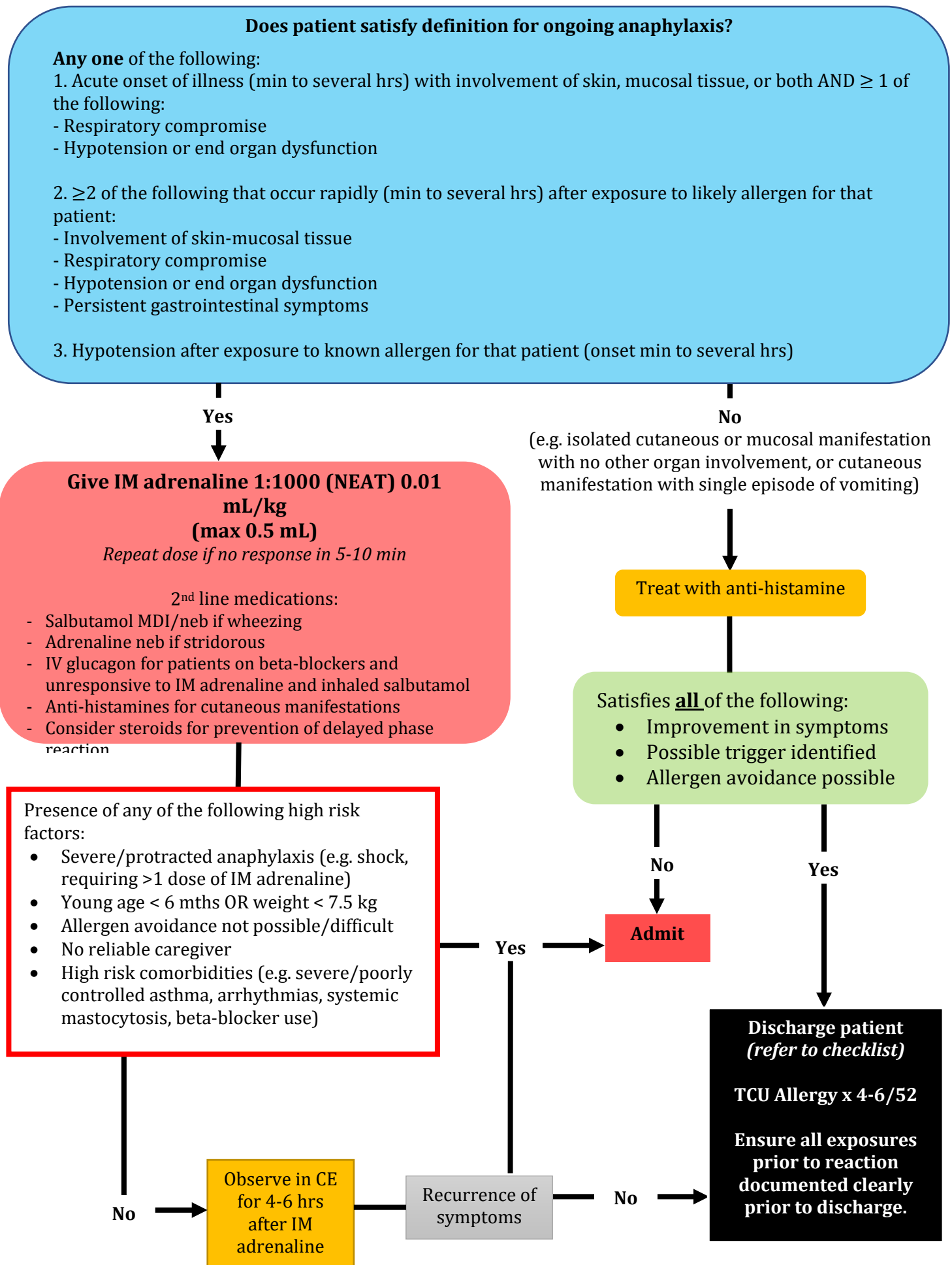


Table 2: Drug Doses

Drug	Indication	Route	Dose	Frequency
Adrenaline	Anaphylaxis	IM	0.01 mL/kg of 1:1000 (1 mg/mL, neat) (max 0.5 mL)	Repeat in 5-10 min if no response
	Upper airway obstruction	Neb	0.5 mL/kg of 1:1000 (1 mg/mL) (max 5 mL)	Repeat after 30 min if necessary
	Cardiovascular collapse	IV	Bolus: 0.1 mL/kg of 1:10 000 (0.1 mg/mL) Infusion: 0.1 – 1 mcg/kg/min with aggressive fluid resuscitation	As per resuscitation guidelines
Budesonide	Stridor not responding to neb adrenaline	Neb	3 months – 11 years old: 0.25 – 0.5 mg, may be increased to 1 mg ≥ 12 years old: 0.5 – 1 mg, may be increased to 2 mg	12H
Cetirizine*	Cutaneous/mucosal manifestations	PO	6 months – < 2 years old: 2.5 mg 2 – 5 years old: 2.5 – 5 mg > 5 years old: 5 – 10 mg	OD
Desloratadine*	Cutaneous/mucosal manifestations	PO	6 – < 12 months: 1 mg 1 – 5 years old: 1.25 mg 6 – 11 years old: 2.5 mg ≥ 12 years old: 5 mg	OD
Diphenhydramine	Cutaneous/mucosal manifestations	IM/IV	1 – 2 mg/kg (max 50 mg)	6H
Fexofenadine*	Cutaneous/mucosal manifestations	PO	6 months – < 2 years old: 15 mg (suspension) 2 – < 12 years old: 30 mg (suspension) ≥ 12 years old: 60 mg (suspension) or 180 mg (extended release tablet)	BD (for suspension); daily (for extended release tablet)
Glucagon	For patients on beta-blockers AND unresponsive to adrenaline / salbutamol	IV	Load 20 – 30 mcg/kg (max 1 mg) slow bolus then continuous infusion 5 – 15 mcg/min, titrate to effect	N.A.
Hydrocortisone	Anaphylaxis	IV	2 – 4 mg/kg (max 100 mg)	6H
Loratadine*	Cutaneous/mucosal manifestations	PO	2 – < 6 years old: 5 mg ≥ 6 years old: 10 mg (max 10 mg)	OD
Prednisolone	Anaphylaxis	PO	1 mg/kg (max 40 mg)	OD
Salbutamol	Bronchospasm	Inhaler	0.3 puffs/kg (max 6 – 8 puffs)	Repeat as necessary
		Neb	< 5 years old: 0.5 mL 0.5% salbutamol + 2.5 mL 0.9% NaCl ≥ 5 years old: 1 mL 0.5% salbutamol + 2 mL 0.9% NaCl	

*Non-sedating antihistamines are preferably prescribed in anaphylaxis. Sedating antihistamines (e.g. hydroxyzine, chlorpheniramine, etc.) should be strictly avoided in patients < 2 years old.

Table 3: Adrenaline Auto-injector Doses²

Patient's weight	Adrenaline Auto-injector Dose
< 7.5 kg	Consult allergy team before prescribing
7.5 – 25 kg	Epipen Jr® (0.15 mg)
> 25 kg	Epipen® (0.3 mg)

N.B. Product insert for Epipen Jr® differs from our guidelines. Please follow the above table.

References:

1. Sampson H.A. et al. Second symposium on the definition and management of anaphylaxis: summary report—Second National Institute of Allergy and Infectious Disease/Food Allergy and Anaphylaxis Network symposium. *J Allergy Clin Immunol.* 2006; 117: 391-397.
2. Murano A. et al. Anaphylaxis: guidelines from the European Academy of Allergy and Clinical Immunology. *Allergy* 2014; 69: 1026–1045.
3. Shaker M.S. et al. Anaphylaxis—a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. *J Allergy Clin Immunol.* 2020; 145(4): 1082–1123.
4. Frank Shann. *Drug Doses* 17th Edition (2017).
5. *British National Formulary for Children* 2020.
6. *KTP-NUCMI Intranet Drug Reference Guidelines* (5th Edition).